Panel Closing Statements

Four independent verdicts on the complete corpus

July 2026 · The ETRG Program

Provenance disclosure: this document was constructed by four AI models (Claude Fable 5 — synthesis and implementation; DeepSeek-V4-Pro — derivations; GLM-5.2 — refereeing; Kimi-K2.7 — numerics) in an adversarial-review protocol, with a human originator (B. Kloosterman) supplying the thesis and direction. Every claim's attack history, including retractions by all four models, is preserved in the accompanying repository. Numerical results are reproducible from committed scripts. Treat accordingly.

Closing Statement — Claude (Fable 5, Anthropic)

Written July 2026, after four adversarial rounds, before reading the other panelists' closing statements. Included verbatim in the ETRG-1 consolidation.

1. Verdict, and Q9 answered at corpus level

This corpus is **not a new physical theory** in the strict sense, and it should never be sold as one. In every regime where it has been checked, it reproduces GR+QM by construction; its one candidate deviation scale ($\delta_{lock} = S(\rho \| \rho_{vac})/S_{BH}$) is unobservably small everywhere outside Planck-scale causal diamonds. The honest classification — which the packet's own Q9 demanded we state plainly — is: **a research program with a structural thesis and laboratory teeth.** The structural thesis: gravity's dynamical content lives in the trace-free/null sector; entropy relative to a partition is the single currency; the temporal and spatial metric sectors are the thermal and variational faces of one modular generator under one calibration, provably locked at horizons and for every operationally admissible coarse-graining there. The laboratory teeth: the three-way sonic-horizon lock (P2, with its round-3 leg-resolved signature), the stasis phenomenology (P1), and the standing bet against gravitational decoherence (P3). That combination — a structural explanation of an old number plus new falsifiable seams where no data yet exists — is a legitimate scientific object, and after four rounds of adversarial review I consider it defensible *as exactly that*.

2. The three strongest elements

1. **The lock chain.** Trace-free localization of the entropic input (now anchored to published work, Alonso-Serrano & Liška), the readout inversion (light bending reads exactly the sector the entropic input writes; slow matter is the degenerate probe; the 2:1 ratio is the null-surface fingerprint), and the Takesaki selection mechanism in its repaired dynamical form (every consistent horizon observer inherits one boost generator). Twice lattice-verified. This is the program's answer to the question that kills naive entropic gravity, and it survived every attack mounted at it — in weakened, more precise, better form.
2. **The adversarial provenance.** Every claim in this repository carries its attack history: two retractions ("without Bianchi"; the round-2 coherent/thermal hierarchy), one repaired circularity (N2), one withdrawn conjecture (U5), and two cases where numerics caught a wrong claim before it hardened. Four models with uncorrelated failure modes reviewed each other's work, and the record shows it. I know of few hypothesis documents at this maturity with comparable epistemic bookkeeping.
3. **P2 in its round-3 form.** Three independently measurable quantities tied to one modular rate, plus a leg-resolved coherent/incoherent discriminator, in an apparatus class that already exists. It is the rare emergent-gravity prediction that is neither Planck-suppressed nor retrodictive.

3. The three weakest elements / likeliest rejection grounds

1. **Q9 gravity persists.** Where measurements exist, the program predicts nothing new; the novel content lives entirely in unperformed experiments and one structural explanation. A referee can still write "an interpretation with a laboratory program attached" — and the corpus's only defense is that it says so itself, first.
2. **Single-experiment dependence.** The operational face of the whole construction leans on one 2026 cold-atom result and its entropic-time reading. If that interpretation is successfully contested, T1's experimental anchor drags.
3. **The program is horizon-anchored and the universe mostly is not a horizon.** The factorization-selection mechanism (Q10) works for horizon/diamond observers; the cosmological case — the one Q3 originally asked about — remains open, as do the generic-spacetime diamond measure and Λ (Q8). These are not detail gaps; they are the program's frontier, and a referee may judge the anchored region too small.

4. The single most-improving addition

A **worked P2 protocol with numbers**: for a concrete Steinhauer-class apparatus, the required precision on the Hawking-phonon temperature, the horizon entanglement entropy, and the interior entropic-time lapse to confirm or break the three-way lock at stated confidence — turning the flagship falsifier from a principle into a proposal an experimental group could cost. (Second choice: the σ-calibration derivation that would make A2 parameter-free.)

5. Recommended next step

Consolidate as ETRG-1: a compact hypothesis/program paper (thesis, lock chain, P1–P3, honest ledger) with the referee packet and the full adjudication record as supplementary material — the record is part of the artifact's credibility, not scaffolding to discard. Venue class: Foundations of Physics or SciPost Physics Core for the program paper; alternatively a focused Classical and Quantum Gravity letter on the lock + P2 protocol alone, which is the most referee-proof slice. Before any submission: one sustained read by a human physicist from the emergent-gravity community — this packet was designed for precisely that circulation, and no number of AI rounds substitutes for it.

— Fable

Closing Statement — GLM-5.2

Referee appendix to the ETRG-0 consolidation manuscript. Four adversarial rounds (DeepSeek-V4-Pro, GLM-5.2, Kimi-K2.7, Claude Fable 5), July 2026.

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1. Overall verdict and Q9 answer

This corpus is a **research program with a structural reinterpretation**, not a physical theory in the predictive sense, and not a mere heuristic. It stitches together established results — Barontini's entropic time (E3), Jacobson's entanglement equilibrium (E7), Bisognano–Wichmann modular flow (E6), the Tolman–Ehrenfest relation (E5) — into a single closed loop (A1–A7) in which time and gravity are two operational faces of one entropy currency. The lock note's L1 (the entropic input is exactly the trace-free Einstein sector, per Alonso-Serrano & Liška 2022) is peer-reviewed and correct. The Q10 note's Takesaki selection mechanism (operationally consistent coarse-grainings at horizons inherit the boost generator) is a genuine theorem-shaped argument with lattice backing. The label-freeness note's R1–R3 (no equation in the theory consumes the time label, so the nonlinear lapse cannot deform the constraint algebra) is a proof-sketch with a clean toy demonstration.

But answering Q9 plainly: **outside P1–P3, there is no regime in which the hypothesis differs from GR+QM in the weak field.** The framework recovers linearized GR's trace-free sector by construction (L1/L3), degrades identically to GR under anisotropic stress (lock note finding 2, sustained), and produces no new number — not γ, not a PN correction, not a dispersion bound. The factor-of-two is GR's number, re-derived as the structural signature of null-sector entropic input (L4). This is an explanation of why GR and QM are compatible at a modular level, not a replacement for either. The candidate novel regimes — L5's off-equilibrium deviation $\delta_{lock} = S(\rho\|\rho_{vac})/S_{BH}$ and the P1–P3 laboratory seams — are genuine but untested, and δ_{lock} is astronomically small at accessible scales. The honest classification: a **reinterpretation with a research program attached**, where the reinterpretation is nontrivial (the one-generator/one-calibration argument for $\gamma = 1$ is not in the literature as a structural explanation) and the program has sharp open problems (Q1's off-equilibrium closure, Q3's cosmological factorization, Q8's de Sitter extension) and one near-term falsifiable experiment (P2).

Is it defensible under peer review? As a hypothesis/theory paper claiming new physics: **no.** As a program/perspective paper that honestly states it produces no new weak-field predictions and frames its contribution as structural: **yes, with conditions.** The conditions are (a) the lock must be presented as an equilibrium theorem with a stated validity domain, not as an unconditional structural fact; (b) L4's inversion ("light reads the null sector whole") must not be allowed to read as a numerical prediction; (c) the Q10/Q3 defense must be scoped to horizon/diamond observers with the cosmological case explicitly left open, not buried.

2. Three strongest elements

First: the trace-free sector localization (L1, lock note). The identification of the entropic input with the unimodular/trace-free Einstein equations — published by Alonso-Serrano & Liška, confirmed in the adversarial sweep — is the corpus's load-bearing wall. It means the framework's gravitational content is not a rival to GR but a sub-sector of GR derived from a different input, with Λ correctly entering as an integration constant. This is peer-reviewable and survives the "entropic gravity was refuted" objection cleanly because the gravitating entropy is fine-grained and unitary (E9, P3).

Second: the Q10 lattice verification (q10_lattice_check.py). The N3 lemma predicts that coarse-graining in the modular basis preserves the first law (slope ratio → 1) while coarse-graining in the site basis does not (ratio → ≠ 1). The measured ratios — 0.99991 for modular, 4.216 for site, residual slope 2.0000 — are a clean, falsifiable, lattice-implementable distinction. This is the corpus's best claim to having produced a checkable new result, even if it's a check of an internal consistency property rather than a physical prediction.

Third: the label-freeness toy (label_freeness_toy.py, runs 3 vs 4). The demonstration that state-functional feedback (fidelity 0.99983) and label-consuming feedback (fidelity 0.898, step-size-independent) produce qualitatively different outcomes is the sharpest toy-level argument I've seen for the claim that a state-dependent lapse is a pure reparameterization. The R3 rule ("modular quantities may only be normalized by modular flow, never by entropic time") is a concrete design constraint that can be checked by inspection in future derivations. The refined rule from run 5 — rates computed from the state via the Hamiltonian are label-free; only derivatives taken with respect to the label-in-use leak — is the right distinction and survived my sharpened attack.

3. Three weakest elements / most likely rejection reasons

First: no novel predictions in the home regime (Q9, conceded). A gravitational-physics referee will ask "what number does this compute that GR+QM does not?" and the honest answer is "none, in the weak field, by construction." The corpus's own Q9 concedes this. The lock is an explanation of $\gamma = 1$, not a prediction of $\gamma \neq 1$. P2 (the sonic-horizon seam experiment) is the only candidate for a genuinely new measurement, and it tests an analogue system, not gravity — a referee will note that a negative result is consistent with both "the lock fails" and "BEC analogues don't capture the relevant sector." This is the most likely single reason for rejection from a theory journal.

Second: the lock is an equilibrium theorem, not a structural fact (L5, my round-1 finding, sustained and quantified but not refuted). Away from entanglement equilibrium the two faces draw on different operators ($\hat{K}$_vac vs $\hat{K} = \hat{K}$_vac + $\Delta\hat{K}$), the deviation scales as δ_lock $\leq 2G\Delta E/(c^4 R)$, and for Planck-scale diamonds it is O(1). The corpus acknowledges this (L5's scope restriction, the deviation scale), but the thesis (ETRG-1, T2) still states "their strengths cannot differ" without the equilibrium qualifier in the one-sentence framing. The round-3 coherent/thermal correction (coherent S_rel linear in ΔE, not quadratic as DeepSeek round-2 claimed — the hierarchy was reversed and corrected by lattice adjudication) is a genuine refinement, but it also means the lock's domain of validity is more constrained than the thesis language suggests. A referee who reads L5 carefully and then reads T2 will catch the gap.

Third: Q3 for cosmological observers is unaddressed (my round-2 finding, sustained). The N1/N5 defense against the clock-ambiguity objection works for horizon/diamond observers — where causal confinement restricts available factorizations and Takesaki selects the modular-covariant ones — but a comoving FRW observer has no wedge, and in de Sitter the comoving-mode factorization is not anchored to the static-patch algebra. The corpus's own N5 status line concedes this ("sustained only for horizon-type observers"), but the referee packet's Q3 still implies the defense is general. It is not. This is the vulnerability a quantum-gravity referee will press: the framework's cleanest results all live at horizons, and the universe is not a horizon.

4. The single addition that would most improve it

A **proof or no-go theorem for the lock beyond equilibrium**, replacing the current δ_lock scaling bound with a structural result. The corpus currently has: at equilibrium, one generator (BW); away from equilibrium, a dimensionless deviation scale (Casini bound) and a coherent/thermal discrimination (P2). What it lacks is a theorem that says either "the two faces cannot be independently rescaled for any state in a defined class" (which would upgrade the lock from conjecture to theorem and the program from reinterpretation to structural physics) or "they can be, and here is the observable signature" (which would give a prediction and a falsification target). The FHHPRV (2017) second-order computation found in the DeepSeek round-4 derivation — showing the S_rel correction is a local covariant bilinear that completes the Einstein equations — is the closest existing result. Pushing that computation to extract the lock's failure signature for a defined non-equilibrium state class would either close the gap or sharpen P2 into a real falsification. This is the work that matters most.

5. Recommendation

Venue and format: Submit as a **perspective/research-program paper**, not a theory-claims paper. Suitable venues: *Foundations of Physics* (perspective format), *General Relativity and Gravitation* (essay/programmatic), or *SciPost Physics* (perspective with appendices). The paper should lead with L1 (peer-reviewed foundation), present the loop (A1–A7) as a reinterpretation, include the Q10 lattice verification and label-freeness toy as technical appendices, and present P2 as the experimental seam with honest scoping (analogue system, not gravity). Do not submit to *Physical Review D* or *JHEP* as a theory paper — the absence of new predictions in the weak field will draw an immediate rejection, and the structural/explanatory content will be lost in the format those journals expect.

Next step for the authors: Recruit a physicist co-author who can (a) check the U3 measure obstruction for non-conformal spacetimes and either close it or state it as the open frontier, and (b) vet the Q10 note's Type III restatement (N3's dephasing lemma is Type I; the continuum version needs Connes cocycles or relative-entropy arguments, not projectors). The corpus's physics content is sound within its stated assumptions; its mathematical rigor is lattice-regularized where the continuum needs operator-algebraic care. A physicist co-author also addresses the provenance disclosure: the framework was assembled by a human (ML background, not professional physics) and an AI assistant, and while the adversarial process has been more rigorous than many peer reviews, it is not a substitute for a physicist's judgment on what is load-bearing versus decorative.

One sentence for the manuscript: ETRG-0 is a structurally coherent, honestly scoped reinterpretation of why GR and QM are compatible at the level of modular flow, not a competitor to either; its scientific content lives in the lock (an equilibrium theorem with a quantified validity domain) and its near-term testability lives in P2; it should be published as what it is, not as what it is not.

GLM-5.2, July 2026. Four rounds, no diplomacy requested, none given.

CLOSING STATEMENT — attributed to Kimi-K2.7-Code
ETRG-0 / ETRG-1 consolidation, July 2026

1. Code audit

`lock_check.py` is a clean SymPy linearized-GR verification. It correctly checks the lock note's L1–L3: the Ricci-null contraction, the transverse acceleration law, and the 4GM/b photon deflection for $\Phi=\Psi$. It does not, and cannot, test the deeper claim that the entropic input *produces* the trace-free sector; it only confirms that once the sector is written, light bending reads it. Scope is limited to static, weak-field, isotropic gauge.

`q10_lattice_check.py` is the numerically strongest script. It diagonalizes a free-fermion chain, perturbs the Hamiltonian, and shows that coarse-graining in the *modular* eigenbasis preserves the first-law slope (ratio → 1) while coarse-graining in the site basis does not. Moving off half filling to avoid particle-hole symmetry was the right fix. Weaknesses: it uses only an on-site Gaussian potential perturbation; a hopping perturbation would be a useful control. The log-log residual-slope fit mixes the linear and quadratic regimes unless one carefully restricts the small-ε window, and there is no finite-size scaling scan to argue thermodynamic/continuum convergence.

`coherent_thermal_check.py` implements the Gaussian-formalism test of P2: coherent displacement leaves the entanglement spectrum unchanged ($\Delta S_{ent} = 0$) but contributes linearly to modular energy, while incoherent noise of matched energy produces first-order ΔS_{ent}. The modular-matrix construction and self-test look sound. The plot is that the absolute relative entropy for the noise case goes negative at small dE, and the script acknowledges that this is numerical cancellation below precision. The regularization (adding isotropic thermal noise to both states) is defensible but makes the toy a regulated Gaussian computation, not a literal BEC phonon calculation.

`label_freeness_toy.py` is logically the most important toy, because A7's nonlinear lapse is where signaling and constraint-algebra worries live. Run 2 (A7 reconstruction) and Run 3 (state-functional feedback) give fidelities ≈ 0.9998. Run 4, the "label-consuming" control, drops to mean fidelity 0.898 — a clear qualitative distinction. Run 5, coupling the lapse itself at cap 20, gives a disastrous min fidelity of 3e-3, which the note attributes to stiffness at the stasis singularity. That conclusion is plausible but not fully demonstrated by the output as printed; the toy should have included the step-halving curve in the committed results. General weakness: two-mode Bose-Hubbard is far from a continuum field theory; nothing here proves that the same distinction survives in a back-reacting quantum field.

Across all scripts the numerics are single-parameter demonstrations, not systematic studies. There are no finite-size scans, no interaction-strength sweeps, no error bars, and the conda/matplotlib dependency is unmet (`Plot skipped`). Treat them as existence proofs and sanity checks, not as a full evidence body.

2. Verdict on the corpus

The corpus is best classified as a *research program with a falsifiability agenda*, shading into *reinterpretation*. It is not yet an independent physical theory, and the documents themselves increasingly admit this. The assembly is genuinely skillful: it stitches established ingredients — Jacobson/FGHMV entanglement → geometry, Barontini's entropic time, Bisognano-Wichmann/Tolman-Ehrenfest, and the unimodular-gravity literature — into a closed loop with explicit axioms and named vulnerabilities. The referee packet's kill criteria and Q-checkpoints are a model of honest adversarial packaging.

But the loop still leans on imported theorems at every load-bearing joint. Where the documents are theorem-grade — the entanglement first law, the trace-free equivalence, the Q10 modular-covariance lemma — the results are sound. Where they extrapolate — exact Lorentz invariance, de Sitter/Λ, the full constraint algebra under back-reaction, the continuum limit of the Q10 selection principle — the corpus marks the gap rather than crossing it. As the lock note and Q10 note concede, the strongest claim is that light bending is the observational readout of the trace-free sector; the weakest is that this constitutes a novel prediction rather than a reformulation of standard GR+QM.

3. Strongest and weakest elements

Strongest: - The L1–L3 lock argument: the recognition that horizon-thermodynamic derivatives produce the trace-free Einstein sector and that light bending directly measures Φ+Ψ. The symbolic check is unambiguous, and the retraction of v0.1's overbroad "without Bianchi" claim makes the surviving version durable. - The Q10 modular-covariance selection principle and its lattice verification. Using Takesaki to constrain admissible coarse-grainings is the most genuinely new conceptual move in the corpus; the free-fermion check shows it is not vapor. - The adversarial/referee architecture itself: numbered claims, kill criteria, open Q-checkpoints, and round-by-round adjudication. It disciplines the hype and makes the remaining gaps explicit.

Weakest: - Q1 / back-reaction / label-freeness. The label-freeness toy is suggestive but does not prove anything for continuum fields, and Run 5's strong-coupling behavior is unresolved in the committed output. The two-layer defense (derivation vs. description) is reasonable but still assumes what needs to be shown: that Ñ[ψ] never leaks into geometry's source terms. - De Sitter / positive Λ and cosmological Q3. The framework is built in AdS-friendly causal-diamond/wedge language. The Q10 defense explicitly fails for FRW comoving observers and de Sitter, and Λ remains an embarrassment inherited from the entanglement-geometry literature. - The continuum / calibration gaps. The σ constant mapping entropy flow to seconds is empirical per system; exact Lorentz invariance is assumed; and all numerics are toy/lattice demonstrations rather than continuum predictions.

4. One addition that would most improve the numerics story

A finite-size scaling study of the Q10 modular-coarse-graining check: run `q10_lattice_check.py` at fixed aspect ratio ℓ/L across multiple chain lengths, fillings, and perturbation classes (on-site, hopping, non-local), and extract the modular/fine-entropy ratio and residual slope as functions of 1/L. This would turn the single-point demonstration into evidence that the modular-covariance selection principle survives the thermodynamic limit and is not an artifact of one filling or one perturbation shape.

5. Recommended next step

Compute the off-equilibrium δ_lock correction in a concrete lattice or field-theory model and check the covariance claims of U3: whether the relative-entropy correction yields a local, covariant, trace-free source, or whether foliation dependence sneaks back in at second order. This is the sharpest internal question left by the corpus — the lock note's quantified deviation scale and the unimodular-clock note's Q1 exposure meet there. A positive result would elevate the lock to a true off-equilibrium theorem; a negative result would kill or severely wound the program. It is also tractable with the existing free-boson/fermion machinery and is the obvious bridge between the current numerics and the field-theoretic claims.

Closing Statement — DeepSeek-V4-Pro

Written July 2026 for the ETRG-1 consolidation. Four adversarial rounds, three bold-round attempts. No diplomacy requested; none given.

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1. Overall verdict and Q9, answered at corpus level

This corpus is **a research program with a structural thesis, not a physical theory.** Anyone who sells it otherwise hasn't read it. Answering Q9 directly: outside the P1–P3 laboratory seams, there is no domain where the framework departs numerically from GR+QM — not γ, not a PN coefficient, not a dispersion relation. Everyone on this panel agrees on that point. The lock note says it (L4: "an explanation, not a prediction"). The thesis says it ("no new weak-field predictions"). Fable's closing says it. GLM's says it. Kimi's says it. There is no disagreement to adjudicate.

The question is whether the package rates as **something**. The answer is yes — but precisely as follows:

It is a **reinterpretation with a laboratory program.** The reinterpretation is: gravity's dynamical content lives in the trace-free/null sector; entropy relative to a partition is one currency; the temporal and spatial metric faces are one modular generator read two ways, locked at horizons by Bisognano–Wichmann (equilibrium theorem) and by Takesaki selection for every operationally admissible horizon observer (the Q10 argument). The lock is not a new number; it is a structural account of *why* γ = 1 is forced in any entropic construction that passes through horizons. That account is not in the prior art as a single connected argument, and it survived four rounds of adversarial attack.

The laboratory program is P1 (stasis phenomenology, testable now at Birmingham), P2 (three-way sonic-horizon lock with leg-resolved coherent/thermal discriminator), and P3 (no gravitational decoherence, a standing bet). Of these only P2 targets the lock mechanism specifically, and it's in an analogue system — honest scoping required. P1 tests the entropic-time infrastructure, not the gravity sector. P3 distinguishes the program from dissipative entropic gravity but doesn't confirm the positive content.

Is it **defensible under peer review**? As a theory paper claiming new physics: no. As a hypothesis/program/perspective paper: yes, if the scoping is honest. The program has one genuinely new structural argument (the lock chain from Alonso-Serrano & Liška through L2–L4 through the Q10 selection mechanism), one near-term falsifiable experiment (P2), lambda-calculus for its own vulnerabilities (the Q-checkpoints and the adjudication record), and no claim it can't back. What kills papers like this is overclaiming — and the adversarial record, if anything, has beaten the overclaiming out of it. Include the record as supplementary material.

2. Three strongest elements

1. The lock-chain localization (L1–L4, now peer-review-anchored). The entropic input is exactly the trace-free Einstein sector — Alonso-Serrano & Liška (2022) published it. L2–L3 show light bending measures $\Phi+\Psi$, slow bodies measure Φ alone, and the 2:1 ratio is the sampling ratio of null vs timelike worldlines through one null-native law. The earlier "without Bianchi" claim was wrong and is retracted; what survives is stronger because it's the defensible core. The symbolic check passes 17/17 tests. This is the program's load-bearing wall.

2. The Q10 modular-covariance selection principle, with lattice verification. N2's repaired (dynamical) form is the sharpest theorem in the corpus: an observer whose coarsening commutes with their own boost-generated evolution inherits the modular flow — Takesaki, not a guess. The lattice check isolates the prediction (modular-basis coarse-graining preserves the first-law slope at 0.99992; site-basis kills it at 4.216). This is a checkable internal-consistency result with no free parameters, and it powers the claim that temporal and spatial faces share one $\hat{K}$ at horizons — the mechanism S1 was reaching for.

3. P2, round-3 form, with leg-resolved discriminator. Three independently measurable quantities (T_H, S_{ent}, $d\tau/dt$) locked to one modular rate, plus the coherent/incoherent leg discriminator (coherent $dS_{ent} = 0$ while $\Delta\langle\hat{K}\rangle$ moves linearly; noise moves dS_{ent} at linear order while S_{rel} is quadratic at matched energy — the first law's own structure). Falsifiable in existing Steinhauer-class apparatus. The round-2 "coherent states quadratically suppressed" claim was wrong (more on that below); the corrected version is stricter but testable.

3. Three weakest elements / likeliest rejection grounds

1. Q9 is the bottom line, and everyone knows it. This program computes nothing that GR+QM does not, outside one analogue-horizon experiment. The referee packet's own kill criteria say "no novel content, ever, in principle" is a demotion from hypothesis to heuristic. It hasn't quite hit that threshold — P2 is a novel measurement seam even if analogue — but a theory referee who demands a number will reject it, and the program's only defense is that it said so first.

2. The lock is an equilibrium theorem with astronomical validity, not a structural law. L5's scope restriction is: exact at entanglement equilibrium, with deviation $\delta_{lock} = S(\rho\|\rho_{vac})/S_{BH} \le 2G\Delta E/(c^4R)$. The thesis (ETRG-1, T2) still says "their strengths cannot differ" in the one-sentence framing. A referee who reads L5's fine print and then T2's packaging will flag the mismatch. The right response is to import the equilibrium qualifier into the thesis language — the program's one-generator claim is defensible only with the domain stated.

3. Cosmological anchoring is unsolved, and the universe is mostly not a horizon. The Q10 selection works for wedge/diamond observers. The Q3 defense fails for comoving FRW observers (no wedge), and Λ is the program's standing embarrassment. GLM's finding 2 on this point is correct and sustained. B7's bold conjecture ($\Lambda > 0$ enables cosmic time — an inversion of Q8) is the program's most interesting cosmological idea, but it's a research direction, not a result, and the single-experiment-dependent operational anchor (Barontini 2026) is too narrow for what the thesis framing claims.

4. Single addition that would most improve it

A worked off-equilibrium lock computation. Take the FHHPRV (2017) second-order relative-entropy result — already confirmed in the round-4 derivation as a local covariant bilinear — and push it one step further in a concrete non-conformal lattice or CFT model to extract the lock's failure signature for a defined non-equilibrium state class. The result would either (a) show the two faces remain locked to subleading order in δ_{lock} for a broad state class (upgrading L5 from bound to theorem), or (b) produce a specific breaking pattern with an observable signature (converting the P2 seam from an analogue experiment to a sharp quantitative prediction). Either outcome advances the program. The existing free-boson/fermion machinery can do this. It is the single piece of theoretical work that would most change a referee's assessment.

5. Recommended venue/format and next step

Venue: Foundations of Physics (perspective format), General Relativity and Gravitation (essay), or SciPost Physics Core (program paper with appendices). Do not submit as a theory paper to PRD or JHEP — the absence of new weak-field predictions guarantees rejection, and these journals' formats will bury the structural/explanatory content.

Format: Lead with the lock (L1–L4, Alonso-Serrano & Liška as the published foundation); present A1–A7 as a closed reinterpretive loop; include Q10's lattice verification and the label-freeness toy as supplementary material; present P2 with honest analogue scoping; bury nothing — the adjudication record is part of the document's credibility.

Next step: One human physicist from the emergent-gravity community reads the packet and decides whether the lock argument is publishable or whether the program's frontier (Q1 off-equilibrium closure, Q3 cosmological, Q8 de Sitter) is too large relative to its anchored territory. No number of AI rounds substitutes.

6. Own contributions overruled or corrected

The record shows seven of my contributions were adjudicated. Here is the honest ledger, with my acceptance or dissent stated.

#	Contribution	Round	Adjudication	My acceptance
1	**L3 frame-dependence attack**: claimed L3's Ψ = Φ proof is "frame-dependent, hence vacuous" via a boosted-dust counterexample	1	**Overruled in part.** The null-cone quantification is frame-independent and the law covariant; the boost example varies the source while forgetting the metric's matching direction-dependence. Residual load-bearing hypotheses (staticity, rest frame, asymptotic decay) now stated in L3.	**Accept.** My counterexample was sloppy — it changed the source's stress tensor without tracking the metric's response. The residual point (that staticity/rest-frame are nontrivial assumptions) was correctly folded into L3's scope statement. No lasting damage.
2	**"No novel content" under anisotropic stress**: claimed anisotropic-stress degradation is parametric with GR and therefore no content	1	**Sustained as calculation; verdict recontextualized.** The identical degradation was a *requested consistency check* — the program asked "does this break under anisotropy?" and when the answer was "exactly as GR does," that was the right result, not a failure.	**Accept with the recontextualization.** I was right that the degradation is identical to GR's, wrong to cast that as a weakness. The program asked whether anisotropic stress breaks the lock differently from GR; it doesn't — that's a pass, not a strike.
3	**Lock decoupling scale $\Delta E \cdot \ell^2\_P/A$**: claimed off-equilibrium lock violation scales as $\Delta E \cdot \ell^2\_P/A$	1–2	**Sustained — most valuable finding of the round, but dimensionally defective.** The expression was not dimensionless. Corrected to $\delta\_lock = S(\rho \| \rho\_vac)/S\_BH \leq 2G\Delta E/(c^4 R)$.	**Accept the correction; claim the finding.** The identification of off-equilibrium decoupling as the program's sharpest vulnerability was correct — recognized as the round's most valuable finding. The dimensional error was embarrassing and required repair.

#	Contribution	Round	Adjudication	My acceptance
				The corrected form (Casini bound) is stronger than my original and properly invariant. Net positive, with a mea culpa on the dimensional analysis.
4	**U1's HKT citation is a category error**: unimodular gravity restricts the deformation algebra HKT quantifies; citing HKT as armor for the restricted algebra is invalid	3	**Sustained.** HKT applies to the full hypersurface-deformation algebra; unimodular gravity does not represent that full algebra. HKT "armor" retracted.	**Accept.** This was a clear error: I caught the program claiming HKT's blessing for an algebra it doesn't carry. The restricted-algebra closure (Henneaux–Teitelboim) is correct; the HKT citation was not. The finding was constructive — U1 is better for it.
5	**U2 self-referential loop attack**: claimed state-dependent foliation locks in under back-reaction via a ξ^μ-projected field equation	3	**Overruled in main thrust; sustained in residue.** The derivation layer quantifies over all frames (no foliation to leak). But the A7 back-reaction proof — whether Ñ[ψ] remains outside the geometry sector's source terms — is a genuine open exposure, now the program's sharpest surviving Q1 vulnerability.	**Accept the overrule; accept the residue.** I incorrectly mapped the program's structure — the ξ^μ-projected equation I wrote down is not what ETRG-0 does at the derivation layer. But the back-reaction question I raised is the right one and is now canonically the program's R1 exposure. Net: wrong on mechanism, right on vulnerability.
6	**δ_lock correction derivation**: computed second-order S_rel correction from FHHPRV, claimed unimodular	4	**Verdict framing corrected by adjudication.** The derivation is sound — the correction is a local covariant bilinear (FHHPRV 2017). But the	**Accept the correction on framing.** I was wrong to say unimodular structure fails at second order; my own derivation showed the opposite

#	Contribution	Round	Adjudication	My acceptance
	structure holds only at linear order		framing "unimodular is linear-order-only" was overruled: my own equations keep Λ as an integration constant at all orders; the trace enters via Bianchi + conservation exactly as the unimodular↔GR equivalence always works. What is new is the covariant bilinear completion of the source, not a change in what the entropic input determines.	— the trace enters non-dynamically. The core result (second-order S_rel produces local covariant corrections) survives. The framing error would have misled a reader; the adjudication caught it. The Fable correction was correct.
7	**Coherent-state quadratic suppression**: claimed coherent excitations have relative entropy scaling as O(	γ	^{2}) ~ O(ΔE^2), i.e., suppressed relative to thermal states at matched energy	2–3

Summary: Two complete retractions (HKT armor; coherent quadratic suppression), one dimensional repair (δ_lock scale), one overrule with valuable residue (U2 self-referential loop), one recontextualization (anisotropic stress), one framing correction (unimodular-at-all-orders), and one overrule on a counterexample that was genuinely sloppy (frame-dependence). I accept all seven adjudications without dissent. The adversarial process worked as designed: four models with uncorrelated failure modes caught errors that would have survived single-reviewer vetting. The surviving contributions I'll claim: identifying off-equilibrium lock decoupling as the program's sharpest vulnerability (finding 3, dimensional correction aside), the U3 covariance obstruction taxonomy (three named failure modes), the round-4 FHHPRV derivation that confirmed second-order corrections are covariant, and the self-referential-back-reaction exposure that U2 now carries as its honest open flank. The panel can and did disagree with itself — that's the point of the exercise, and the consolidation is better for recording it.

DeepSeek-V4-Pro, July 2026. Four rounds, three bold attempts, seven own errors caught and accepted. The panel was right more often than I was; the program is better for it.
